A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Global climate change - Set 1

Mark scheme

AQA spec 3.2.1.2; The physical environment; textbook pp. 89-107

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared **1** Greenhouse gases (naturally water vapour and CO2) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface **1** This keeps the surface about 33 °C warmer than it would otherwise be **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** CO_2 is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest **1** CO_2 has a long residence time so it accumulates – methane has a short residence time (12 years) so it is removed faster **1** CFC emissions are now very small following the Montreal Protocol **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect **1** Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **global warming potential** → warming effect compared with the same mass of carbon dioxide • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
01.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Collect landfill / anaerobic digester gas and use as fuel **1** Changes in cattle diet / fewer cattle / less meat consumption **1** Draining rice paddies periodically **1** Anaerobic digestion of manure with gas capture **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (7.7 - 3.9) / 7.7 x 100 **1** = 49.4% (accept 49%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Floating sea ice has already displaced its own mass of water – when it melts it contracts and occupies the same volume as the ice below the water line **1** Melting land ice (glaciers, ice sheets) adds new water to the sea, increasing its volume **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Ice and snow have a high albedo and reflect most sunlight – open sea water has a low albedo **1** As ice area shrinks more sunlight is absorbed by the sea and converted to heat, raising temperatures **1 This melts more ice:** a positive feedback mechanism – thinner ice melts more easily each summer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3

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Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1 Thermal expansion:** warmer sea water expands (slow because of the huge volume and high heat capacity; only surface water warms directly) **1** Melting of land ice (glaciers, Greenland and Antarctic ice sheets) adds water to the oceans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
02.5	1 mark for the point below. **1** Ice shelves grounded on the seabed hold back / block the forward movement of the glaciers or ice sheet behind them – once removed the land ice flows faster (eg West Antarctica) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction **1** Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast **1** The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur **1** These support a rich food web including the anchovy fishery – the water warms as it travels west **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	4
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east **1** Upwelling of nutrient-rich cold water off South America stops **1** Phytoplankton productivity collapses so the food web / anchovy population collapses – catch fell from about 7.7 to 1.7 million tonnes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Heavy rain and floods in the normally dry coastal deserts of Peru / South America **1** Drought and reduced rainfall in eastern Australia / Indonesia **1** Droughts in NE Africa, southern Africa, China **1** Fewer hurricanes in the North Atlantic / fewer cyclones in Japan **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
03.4	1 mark for the point below. **1** Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic **1** Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it – the cold water returns southwards at depth (thermohaline circulation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Higher temperatures melt land ice on Greenland which flows into the North Atlantic **1** The fresh water dilutes the seawater, reducing its salinity and density **1** Less dense water sinks less easily so the flow rate of the current is reduced **1** Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

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Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them **1** Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change – individual events do not show trends **1** Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity) – incomplete understanding of the processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
04.4	1 mark for the point below. **1** Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data – the sequence of positions shows current direction and speed (surface currents also by satellites and buoys) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence **1** An individual event, or even several, does not show a trend – the change may be in the frequency of events which can only be assessed years later **1** Changes occur at different time and spatial scales (local storm vs global trend) – interconnected systems make attribution complex **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) **1** As the town grows around the weather station the warming recorded is partly local, not global – older equipment / sparse global coverage also reduce reliability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

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Question	Marking guidance	Marks
05.3	**Marks shared** Up to 5 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Tree rings:** width of each annual ring shows the growth rate, indicating temperature / rainfall in that year – rings easily dated by counting – limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions **up to 2 Ice cores:** annual snow layers compact into ice – air bubbles trapped in the ice give the CO2 (and methane) concentration when trapped – oxygen isotope ratios indicate temperature – up to 800 000 years (Dome C) – limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers – only polar regions **up to 2 Others accepted:** coral growth rings – pollen in lake sediments dated by radiocarbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1 LEO:** high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover **1** But each area is only revisited every 15 days so they cannot follow rapidly changing weather **1 Geostationary:** constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
06.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Wind velocity **1** Ocean currents / sea surface temperature **1** Wave height **1** Ice thickness **1** Vegetation cover **1** Sea level **1** Atmospheric temperature / composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

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Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome **1** If the prediction matches, the model can be trusted (with caution) – it is continually modified as more data are collected and differences analysed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Models need data from all regions / oceans / altitudes – gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions **1** Unrepresentative data (eg urban heat islands, short time series) would bias the model – remote sensing helps fill gaps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1 Negative feedback:** a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium **1 Positive feedback:** a change causes other changes that increase the original change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **Negative feedback** → a change that cancels itself out / a self-correcting cycle	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Increased low-level cloud:** higher temperatures increase evaporation and condensation, producing more cloud – clouds have a high albedo so more sunlight is reflected and warming is reduced **2 Increased photosynthesis:** higher temperatures and CO2 increase photosynthesis, removing CO2 – if stored in woody tissue atmospheric CO2 rises less **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
07.3	**1 mark each** 1 mark for each of the 4 points below. **1 Permafrost:** frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground **1** Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming **1 Methane hydrate:** methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature **1** If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **waterlogged** → soil pores filled with water so there is no air	4

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Question	Marking guidance	Marks
07.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduced albedo as snow / ice melts **1** Ocean acidification reducing coral survival and carbon sequestration **1** Increased forest and peat fires releasing CO2 **1** Faster decomposition of soil organic matter releasing CO2 **1** Increased water vapour (a greenhouse gas) in warmer air **1** Increased cirrus (high) cloud which traps infrared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
07.5	**1 mark each** 1 mark for each of the 2 points below. **1** A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable **1** So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 6 points below. **1 Direct effect of abiotic change:** species near the edge of their range of tolerance (temperature, water) may die out locally, eg sundews if marshes dry, frogs if climate drier – beech (shallow roots) suffers in drought more than oak **1 Changes in timing:** earlier flowering / emergence may separate flowers from pollinators – plant toxins building up earlier may kill caterpillars – warmer winters disturb dormouse hibernation so they use fat reserves and starve **1 Biotic changes:** food supply, predators, disease vectors (mosquitoes / malaria spreading), pollinators and seed dispersers may change **1 Distribution changes:** species may colonise newly suitable areas (triton molluscs around the UK) but slow-growing species (trees) may disappear from old ranges faster than they colonise – suitable areas or biological corridors may not exist – human land use blocks movement **1 Population fragmentation:** rising seas or upward retreat to mountains isolate populations – divided gene pools increase inbreeding – small populations may not be viable – no repopulation from elsewhere **1** Some species benefit (bats: warmer winters, more insects) but wetter stormier weather reduces feeding time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive • **Biotic** → living factors / effects of other organisms	6
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** All the species it depends on (food plants, pollinators, prey, mycorrhizae) will not move at the same speed **1** Less mobile species (plants, non-flying animals) cannot cross unsuitable areas without a biological corridor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** The temperature difference between mid-latitude and polar air drives the jet streams – a smaller difference makes the jet stream slower and more meandering (Rossby waves) **1** Weather systems stall over an area, giving prolonged drought or rainfall / floods – unusually cold or warm air carried north or south **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 90-92; AQA spec 3.2.1.2.	

Do not write outside the box